Governing relations of the $CaCO_3$-barrier screen: derivation

Sessions P0-W3.1 and P0-W3.2, contract v2 (symbolic; no values)

Each section opens with the physical argument, and the algebra follows it. Every displayed relation carries its SI units in brackets. The notation is that of `theory/symbols.yaml`, the functions are those of `theory/contract.yaml`, and open choices for the sign audit are listed in `theory/open_questions.md` (cited here as Q1, Q2, ...).

1 Regimes and what each link must achieve

Physics. The proposed mechanism is assembled in one chemical environment and must survive in another. The template is anchored, binds Ca^{2+} and templates the mineral while the electrode is a water-processed, alkaline slurry at room temperature (R-slurry). The finished interface then sits in a non-aqueous, fluorine-free electrolyte with the Al collector held at the cathode's anodic potential (R-cell). The two regimes have different solvents, different reference states and different corrosion modes, so a relation is meaningful only in the regime whose reference states it uses. Every relation below is therefore tagged with its regime, and none is carried across without saying so.

Table 1 states, for each link, the necessary condition the link must meet in its regime. These are directions of inequalities, not thresholds: the P0 thresholds, margins and decision rules are in `theory/thresholds.md`. Meeting them makes the mechanism thermodynamically possible under the stated framing; it does not show that a barrier works, and no rate, capacity or device statement follows from any of them.

Link	Regime	What the link must achieve	Relation
C1 anchor	R-slurry	template held at the oxide against water	$W_{\mathrm{adh,wet}} > 0$, Eq. (1)
C2 sequester	R-slurry	Ca^{2+} held at the template against hydrated Ca^{2+}	$\Delta G_{\mathrm{seq}} < 0$, Eq. (7); it acts on C3 only through $\Delta\gamma_{\mathrm{ads}}$, Eq. (30)
C3 nucleate	R-slurry	heterogeneous barrier lowered, and accessible at Ω_{bulk}	$f(\theta)$ small; $\Delta G^{*}_{\mathrm{het}}(\Omega_{\mathrm{bulk}}) \le B_{\mathrm{acc}}$, Eqs. (22), (24)
C4 conformal	R-slurry, R-cell	a continuous film that closes below the thickness bound	$S_{\mathrm{spread}} \ge 0$ or closure below $d_{\max}$, Eqs. (33), (43)
C5 stable	R-slurry	mineral and oxide on the aqueous Pourbaix hull	$\Delta G_{\mathrm{pbx}} = 0$ (tolerance at G2), Eq. (37)
C5 stable	R-cell	phases inside their Li-referenced window	$E_{\mathrm{red}} < E_{\mathrm{Li}} < E_{\mathrm{ox}}$, Eq. (39)

Table 1: Necessary conditions per link. C1–C3 are defined only in R-slurry: their reference states (water-wetted surfaces, hydrated Ca^{2+}, an aqueous ion activity product) do not exist in the cell.

Two rules follow from the regime split. First, C1–C3 make no statement about the cell: anchor retention, Ca^{2+} binding and nucleation in the organic electrolyte are not computed. Second, C5 needs two different constructions (Section 9), because aqueous Pourbaix analysis assumes water as the solvent and the hydrogen electrode as the reference.

2 Adhesion in water

Physics. In the slurry the template never meets a dry oxide. The hydroxylated Al_2O_3/AlOOH surface is covered by hydrogen-bonded water, and the polymer is hydrated too. A catechol–oxide contact can only form by displacing that water, so the relevant comparison is between two states that actually compete in the slurry: the oxide and polymer in contact, and the oxide and polymer each wetted by water. The dry (vacuum) work of adhesion compares the contact with two bare surfaces, a state that never occurs in R-slurry, and is therefore only a comparator.

The reversible work per unit area to separate a contact between phases A and B inside a medium M is the Dupré work $W = \gamma_{AM} + \gamma_{BM} - \gamma_{AB}$: the two new A–M and B–M interfaces are created and the A–B interface is destroyed. With A the oxide, B the polymer and M water,

$$W_{\mathrm{adh,wet}} = \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]. \tag{1}$$

Units: each term is an interfacial free energy, so $W_{\mathrm{adh,wet}}$ is in $\mathrm{J\,m^{-2}}$. Writing every in-water energy through the dry works of each surface with water, $\gamma_{\mathrm{ow}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{w}} - W_{\mathrm{ow}}$ and $\gamma_{\mathrm{pw}} = \gamma_{\mathrm{p}} + \gamma_{\mathrm{w}} - W_{\mathrm{pw}}$, gives

$$W_{\mathrm{adh,wet}} = W_{\mathrm{adh,dry}} + 2\gamma_{\mathrm{w}} - W_{\mathrm{ow}} - W_{\mathrm{pw}}, \qquad W_{\mathrm{adh,dry}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]. \tag{2}$$

Equation (2) is the water-competition statement in one line: a large dry adhesion is offset by the works W_{ow} and W_{pw} with which water already binds each surface. A hydrophilic, hydroxylated oxide and a hydrophilic polymer bind water strongly, so $W_{\mathrm{adh,wet}}$ can be small or negative even when $W_{\mathrm{adh,dry}}$ is large. The bare-oxide control has $W_{\mathrm{adh,wet}} = 0$ by construction and fixes the zero of the scale.

The interfacial energies of Eq. (1) are not computed separately. The potential of mean force (PMF) of a template fragment along a separation coordinate in explicit water gives the free-energy difference between the separated and bound states, ΔF_{PMF} (J per fragment), which already contains the displacement penalty and the entropy of the released water. Normalising by the contact area,

$$W_{\mathrm{adh,wet}} = \frac{\Delta F_{\mathrm{PMF}}}{A_{\mathrm{c}}} \quad [\mathrm{J\,m^{-2}}], \tag{3}$$

with units $\mathrm{J/m^2}$. Two conditions make Eq. (3) meaningful: the PMF must reach a plateau at large separation (both surfaces fully rewetted), and A_{c} must be defined once for all fragments, because the ranking across binders depends on it (open question Q5). C1 requires $W_{\mathrm{adh,wet}} > 0$ beyond its uncertainty on the populated oxide terminations.

3 The sequestration cycle and the enrichment factor

Physics. The template can raise the local Ca^{2+} concentration only if it holds Ca^{2+} more strongly than water does. The question is a competition between two ligand environments for the same ion: the hydration shell in bulk solution and the coordinating groups of the template. The free energy must therefore start from hydrated Ca^{2+} at the regime activity, include the cost of preparing the binding site at the regime pH, and end in a well-defined local state.

Cycle and standard states. Let the coordinating group be an acid L–H that binds Ca^{2+} only when deprotonated. The cycle is

$$\begin{aligned}
&\text{(i)} \quad \mathrm{L{-}H\ (layer)} \ \rightarrow\ \mathrm{L^-(layer) + H^+(aq)}, && \Delta G_{\mathrm{i}} = -RT\ln\alpha \quad [\mathrm{J\,mol^{-1}}] \\
&\text{(ii)} \quad \mathrm{Ca^{2+}(aq,\ bulk) + L^-} \ \rightarrow\ \mathrm{Ca^{2+}\cdot L^-(layer)}, && \Delta G_{\mathrm{ii}} = \Delta G_{\mathrm{site}} \quad [\mathrm{J\,mol^{-1}}]
\end{aligned}$$

with $\alpha = 1/\left(1 + 10^{\,\mathrm{p}K_{\mathrm{a}} - \mathrm{pH}}\right)$ (dimensionless). Step (i) is the free energy of finding the site deprotonated when the solution fixes the proton activity: α is the deprotonated fraction at the regime pH,

and $-RT\ln\alpha \geq 0$. Step (ii) is the transfer of one Ca^{2+} from bulk solution into the binding layer. Summing,

$$\Delta G_{\mathrm{seq}} = \Delta G_{\mathrm{site}} + RT\ln\bigl(1 + 10^{\,\mathrm{p}K_{\mathrm{a}}-\mathrm{pH}}\bigr) \quad [\mathrm{J\,mol^{-1}}]. \tag{4}$$

Units: RT is in $\mathrm{J\,mol^{-1}}$ and the logarithm is of a pure number. The correction vanishes as $\mathrm{pH} - \mathrm{p}K_{\mathrm{a}} \to \infty$, equals $RT\ln 2$ at $\mathrm{pH} = \mathrm{p}K_{\mathrm{a}}$, and grows as the site becomes protonated. Both $\mathrm{p}K_{\mathrm{a}}$ and the activity coefficients are taken at the regime ionic strength. This is why catechol, largely protonated near neutral pH, must be tested rather than assumed (design flag 4).

Step (ii) fixes the standard states. Let $w(\mathbf{r})$ be the PMF of one Ca^{2+} relative to bulk solution (J per ion, $w \to 0$ far from the template), computed with counter-ions and water explicit. In the dilute limit the equilibrium concentration is $c(\mathbf{r}) = c_{\mathrm{bulk}}\,\mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}$. Assigning to each site a volume V_{site} (the binding-layer volume per site) and defining

$$\Delta G_{\mathrm{site}} \equiv -RT\ln\Bigl[\frac{1}{V_{\mathrm{site}}}\int_{\mathrm{site}} \mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}\,\mathrm{d}^3r\Bigr] \quad [\mathrm{J\,mol^{-1}}] \tag{5}$$

(the bracket is a ratio of volumes, so it is a pure number) makes the transfer free energy a partition between two concentration scales: the bulk at the regime activity on the solution side, and the mean concentration in the site volume on the layer side. The conventional binding free energy to a 1 mol-per-volume standard state c° differs from it by the standard-state correction

$$\Delta G^{\circ}_{\mathrm{bind}} = \Delta G_{\mathrm{site}} - RT\ln\bigl(N_{\mathrm{A}}c^{\circ}V_{\mathrm{site}}\bigr) \quad [\mathrm{J\,mol^{-1}}], \tag{6}$$

where $N_{\mathrm{A}}c^{\circ}V_{\mathrm{site}}$ is dimensionless. A literature binding constant can therefore be compared with ΔG_{site} only after Eq. (6), and only with the same V_{site}. The choice of V_{site} shifts ΔG_{site} by $RT\ln(V'_{\mathrm{site}}/V_{\mathrm{site}})$, so it is fixed once (open question Q4).

Activity versus concentration. Equation (5) defines a concentration ratio. Because w contains every interaction of the ion with its surroundings, the electrochemical potential of Ca^{2+} is uniform at equilibrium (Section 4): the ion in the layer carries the same chemical potential as in bulk, and the whole enrichment sits in its local standard potential, equivalently in a local activity coefficient $\gamma_{\mathrm{loc}} = \gamma_{\mathrm{bulk}}/\mathcal{E}$. $\mathcal{E}$ is therefore a concentration ratio and never an activity ratio. Contract v1 used a *local-ideal* bookkeeping that gave the enriched ions the bulk standard state; Section 7 shows that it is inconsistent at equilibrium and replaces it.

Enrichment and its validity range. With the local-ideal bookkeeping and dilute, non-interacting sites,

$$\mathcal{E} \equiv \frac{c_{\mathrm{local}}}{c_{\mathrm{bulk}}} = \exp\Bigl(-\frac{\Delta G_{\mathrm{seq}}}{RT}\Bigr) \quad [1]. \tag{7}$$

$\Delta G_{\mathrm{seq}} < 0$ gives enrichment ($\mathcal{E} > 1$); $\Delta G_{\mathrm{seq}} > 0$ gives depletion: the template is then a worse ligand than water. Equation (7) holds when (a) Ca^{2+} is dilute in bulk, (b) the sites are far from saturation, (c) one ion binds per site and sites do not interact, and (d) the non-specific (ionic-strength) activity coefficient is the same in the layer as in bulk, so that $\mathcal{E}$ is fixed by the PMF alone. Condition (b) is quantified next.

Saturation bound. The layer has Γ_{s} sites per unit area in a thickness δ_{s}, so its Ca^{2+} content cannot exceed $c_{\max} = \Gamma_{\mathrm{s}}/\delta_{\mathrm{s}}$ ($\mathrm{mol\,m^{-2}}$ over m gives $\mathrm{mol\,m^{-3}}$). For independent, equivalent sites (Langmuir), the occupied fraction is $\Theta = x/(1+x)$ with $x = \mathcal{E}c_{\mathrm{bulk}}/c_{\max}$, and $c_{\mathrm{local}} = c_{\max}\Theta$:

$$c_{\mathrm{local}} = \frac{\mathcal{E}c_{\mathrm{bulk}}}{1 + \mathcal{E}c_{\mathrm{bulk}}/c_{\max}} \leq \min\bigl(\mathcal{E}c_{\mathrm{bulk}},\ c_{\max}\bigr) \quad [\mathrm{mol\,m^{-3}}]. \tag{8}$$

The dilute result (7) is the limit $x \ll 1$; at $x \gg 1$ the enrichment saturates at $c_{\max}/c_{\mathrm{bulk}}$ however negative ΔG_{seq} is. A strongly binding but sparse template is therefore bounded by its site density, not by its binding free energy. Electrostatic coupling between charged sites, which Langmuir ignores, is open question Q3.

Occupancy and surface excess. The same closure gives the fraction of occupied sites and the Gibbs relative excess of Ca^{2+} at the template–liquid interface, the bound amount per area less the amount the layer volume would hold at the bulk concentration:

$$\Theta = \frac{x}{1+x} = \frac{c_{\mathrm{local}}}{c_{\max}}, \qquad x = \frac{\mathcal{E} c_{\mathrm{bulk}}}{c_{\max}} \quad [1], \qquad \Gamma_{\mathrm{Ca}} = \Gamma_{\mathrm{s}}\Theta - \delta_{\mathrm{s}} c_{\mathrm{bulk}} \quad [\mathrm{mol\,m^{-2}}]. \tag{9}$$

x is the occupancy odds of a site at the regime activity. $\Gamma_{\mathrm{Ca}} > 0$ for net binding and $\Gamma_{\mathrm{Ca}} < 0$ for a template that repels Ca^{2+}. These, not a local supersaturation, are what C2 hands to C3 (Section 7).

4 The driving force at the template

Physics. A nucleus is built from ions taken out of the solution, and the free energy it gains per formula unit is the chemical potential those ions carry minus that of the solid. The template sits in an open interfacial region that exchanges ions with the slurry. At equilibrium nothing can be gained by moving an ion between the bulk and the layer, so every ion carries the same electrochemical potential everywhere: the template can concentrate ions, but it cannot make them carry more free energy. The driving force at the template is therefore the bulk one. The algebra below shows this for the two ways a template enriches Ca^{2+}, specific binding and a mean electrostatic potential.

One definition. Throughout,

$$\Omega = \frac{\mathrm{IAP}}{K_{\mathrm{sp}}} = \frac{a_{\mathrm{Ca^{2+}}}\, a_{\mathrm{CO_3^{2-}}}}{K_{\mathrm{sp}}} \quad [1], \qquad \Delta\mu = k_{\mathrm{B}} T \ln \Omega \quad [\mathrm{J}], \tag{10}$$

with K_{sp} the solubility product of the registered framing (ACC or calcite), $\Delta\mu$ the chemical-potential drop per $CaCO_3$ formula unit on crystallisation, and V_{m} the molar volume per $CaCO_3$ formula unit. Units: activities and K_{sp} are pure numbers on the molal scale; $k_{\mathrm{B}}T$ is in J. Nucleation requires $\Omega > 1$. This is the charter's convention (F3): its S is this Ω. The per-ion square root $\Omega^{1/2}$ with a factor $2k_{\mathrm{B}}T$ is the same statement and is not used, so that no factor of two can be lost between lanes.

Equilibrium of the open interfacial region. Write the electrochemical potential of ion i (charge number z_i) at position $\mathbf{r}$ as

$$\tilde{\mu}_i(\mathbf{r}) = \mu_i^{\circ} + w_i^{\mathrm{sp}}(\mathbf{r}) + k_{\mathrm{B}} T \ln \frac{\gamma_i(\mathbf{r})\, m_i(\mathbf{r})}{m^{\circ}} + z_i e \psi(\mathbf{r}) \quad [\mathrm{J}], \tag{11}$$

where μ_i° is the bulk standard potential, w_i^{sp} the specific (non-electrostatic) interaction with the template ($w^{\mathrm{sp}} \to 0$ in bulk), γ_i the non-specific activity coefficient, m_i the molality and ψ the mean electrostatic potential ($\psi \to 0$ in bulk). Equilibrium of the region with the slurry, which acts as a reservoir, is the statement

$$\tilde{\mu}_i(\mathbf{r}) = \tilde{\mu}_{i,\mathrm{bulk}} \quad \text{for every ion } i \text{ and every } \mathbf{r}. \tag{12}$$

Solving Eq. (11) for m_i gives the local enrichment, $m_i(\mathbf{r})/m_{i,\mathrm{bulk}} = (\gamma_{i,\mathrm{bulk}}/\gamma_i) \exp[-(w_i^{\mathrm{sp}} + z_i e\psi)/k_{\mathrm{B}}T]$: the Boltzmann factor of Section 3 is a *consequence* of Eq. (12), not an exception to it.

The driving force for adding one formula unit to a nucleus at $\mathbf{r}$ is

$$\Delta\mu(\mathbf{r}) = \tilde{\mu}_{\mathrm{Ca}}(\mathbf{r}) + \tilde{\mu}_{\mathrm{CO_3}}(\mathbf{r}) - \mu_{\mathrm{CaCO_3(s)}} = \tilde{\mu}_{\mathrm{Ca,bulk}} + \tilde{\mu}_{\mathrm{CO_3,bulk}} - \mu_{\mathrm{CaCO_3(s)}} = k_{\mathrm{B}} T \ln \Omega_{\mathrm{bulk}} \quad [\mathrm{J}]. \tag{13}$$

It does not depend on $\mathbf{r}$: it is the same at a binding site, in the diffuse layer and in the bulk.

(i) Specific binding. At a site, $w_{\mathrm{Ca}}^{\mathrm{sp}}$ adds $\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$ per ion to the standard part of $\tilde{\mu}_{\mathrm{Ca}}$ (a lowering when $\Delta G_{\mathrm{seq}} < 0$) (with the site-volume standard state of Eq. (5)) and raises the concentration term by $k_{\mathrm{B}}T \ln \mathcal{E} = -\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$. The two changes are equal and opposite, which is why $\tilde{\mu}_{\mathrm{Ca}}$ stays at its bulk value. The local enrichment $c_{\mathrm{local}}/c_{\mathrm{bulk}} = \mathcal{E} = \exp(-\Delta G_{\mathrm{seq}}/RT)$ therefore changes the driving force by

$$\Delta\mu_{\mathrm{site}} - \Delta\mu_{\mathrm{bulk}} = \underbrace{k_{\mathrm{B}}T \ln \mathcal{E}}_{\text{concentration}} + \underbrace{\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}}_{\text{standard potential}} = 0 \quad [\mathrm{J}]. \tag{14}$$

(ii) A mean potential. At an anionic template ($\psi < 0$) Ca^{2+} is enriched by $e^{-2e\psi/k_{\mathrm{B}}T}$ and CO_3^{2-} depleted by $e^{+2e\psi/k_{\mathrm{B}}T}$. Even counting concentrations alone, the ion product is unchanged, $(a_{Ca^{2+}}a_{CO_3^{2-}})_{\mathrm{local}} = (a_{Ca^{2+}}a_{CO_3^{2-}})_{\mathrm{bulk}}$, and in Eq. (13) the potential enters as $(z_{\mathrm{Ca}} + z_{\mathrm{CO_3}})e\psi = 0$ because the solid is neutral. A mean potential changes neither Ω nor $\Delta\mu$. This settles Q2: carbonate needs no separate treatment, and holding it at its bulk activity (the v1 choice) is not the issue; crediting the enriched Ca^{2+} was.

Result. At equilibrium the nucleus at the template is driven by $k_{\mathrm{B}}T \ln \Omega_{\mathrm{bulk}}$, evaluated at the regime activities. The template changes the barrier only through the interfacial energies: the diffuse-layer free energy is part of γ_{sl} and γ_{nl}, and the specific binding enters γ_{sl} through the Gibbs adsorption equation (Section 7). The local-ideal ratio $\Omega_{\mathrm{li}} = (c_{\mathrm{local}}/c_{\mathrm{bulk}})\Omega_{\mathrm{bulk}}$, with $\Delta\mu_{\mathrm{li}} = \Delta\mu_{\mathrm{bulk}} - \Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$, survives only as the named double-counting demonstration of the contract (`Omega_local_ideal`, `Delta_mu_local_ideal`).

5 Homogeneous and heterogeneous nucleation

Physics. A cluster of the new phase gains $\Delta\mu$ for every formula unit it contains but pays γ for every unit of interface it creates. Volume grows as r^3 and surface as r^2, so small clusters are unstable and large ones grow; the barrier is the free energy at the crossover. On a substrate the cluster can replace part of its interface with a cheaper substrate contact, which lowers the barrier without changing the driving force.

Homogeneous barrier. In the capillarity approximation, a spherical cluster of radius r with molecular volume $v_{\mathrm{m}} = V_{\mathrm{m}}/N_{\mathrm{A}}$ has

$$\Delta G_{\mathrm{hom}}(r) = -\frac{4\pi r^3}{3}\frac{\Delta\mu}{v_{\mathrm{m}}} + 4\pi r^2\gamma_{\mathrm{nl}} \quad [\mathrm{J}]. \tag{15}$$

Units: r^3/v_{m} counts formula units, times J per formula unit; $r^2\gamma_{\mathrm{nl}}$ is m^2 times $\mathrm{J\,m^{-2}}$. Setting $\partial\Delta G_{\mathrm{hom}}/\partial r = 0$,

$$r^* = \frac{2\gamma_{\mathrm{nl}}v_{\mathrm{m}}}{\Delta\mu} \quad [\mathrm{m}], \qquad \Delta G_{\mathrm{hom}}^* = \frac{16\pi\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\Delta\mu^2} = \frac{4\pi r^{*2}\gamma_{\mathrm{nl}}}{3} \quad [\mathrm{J}]. \tag{16}$$

Units: $\mathrm{J\,m^{-2}}\cdot\mathrm{m^3/J} = \mathrm{m}$ for r^*; $\mathrm{J^3\,m^{-6}}\cdot\mathrm{m^6/J^2} = \mathrm{J}$ for $\Delta G_{\mathrm{hom}}^*$. With $\Delta\mu = k_{\mathrm{B}}T\ln\Omega$, $\Delta G_{\mathrm{hom}}^* \to \infty$ as $\Omega \to 1^+$ (no driving force, no nucleation) and $\Delta G_{\mathrm{hom}}^* \to 0$ as $\Omega \to \infty$; it increases with γ_{nl} and V_{m} and decreases with T at fixed Ω. The absolute barrier in units of $k_{\mathrm{B}}T$ is $\Delta G_{\mathrm{hom}}^*/k_{\mathrm{B}}T$.

Spherical-cap geometry. On a flat substrate the cluster is a cap cut from a sphere of radius r, meeting the substrate at contact angle θ measured through the cap. Elementary geometry gives base radius $\rho = r\sin\theta$, height $h = r(1-\cos\theta)$, and

$$A_{\mathrm{cap}} = 2\pi r^2(1-\cos\theta), \qquad A_{\mathrm{base}} = \pi r^2\sin^2\theta \quad [\mathrm{m}^2], \tag{17}$$

$$V_{\mathrm{cap}} = \frac{\pi r^3}{3}(1-\cos\theta)^2(2+\cos\theta) \quad [\mathrm{m}^3]. \tag{18}$$

The cap's free energy has three parts: the volume term, the free nucleus–liquid surface, and the base, where substrate–liquid interface (γ_{sl}) is replaced by substrate–nucleus interface (γ_{sn}):

$$\Delta G_{\mathrm{het}}(r) = -V_{\mathrm{cap}}\frac{\Delta\mu}{v_{\mathrm{m}}} + \gamma_{\mathrm{nl}}A_{\mathrm{cap}} + (\gamma_{\mathrm{sn}} - \gamma_{\mathrm{sl}})A_{\mathrm{base}} \quad [\mathrm{J}]. \tag{19}$$

Stationarity of Eq. (19) with respect to θ at fixed cap volume is the contact-line balance, Young's relation $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} = \gamma_{\mathrm{nl}}\cos\theta$ (Section 6). Substituting it, the surface terms become

$$\gamma_{\mathrm{nl}}\pi r^2\big[2(1-\cos\theta) - \cos\theta\sin^2\theta\big] = \gamma_{\mathrm{nl}}\pi r^2(1-\cos\theta)^2(2+\cos\theta) = 4\pi r^2\gamma_{\mathrm{nl}}\,f(\theta) \quad [\mathrm{J}], \tag{20}$$

using $2 - \cos\theta - \cos^2\theta = (1-\cos\theta)(2+\cos\theta)$, and the volume term is $\frac{4\pi r^3}{3}f(\theta)$ times $-\Delta\mu/v_{\mathrm{m}}$, with

$$f(\theta) = \frac{(2+\cos\theta)(1-\cos\theta)^2}{4} = \frac{2 - 3\cos\theta + \cos^3\theta}{4} \quad [1]. \tag{21}$$

Hence $\Delta G_{\mathrm{het}}(r) = f(\theta)\,\Delta G_{\mathrm{hom}}(r)$ at every r: the cap has the same critical radius as the sphere, and

$$\Delta G^*_{\mathrm{het}} = f(\theta)\,\Delta G^*_{\mathrm{hom}} \quad [\mathrm{J}]. \tag{22}$$

The factorisation holds only because one and the same γ_{nl} multiplies A_{cap} and enters Young's relation; this is the reason Section 6 keeps the bare γ_{nl} in Young.

Properties of f. $f(0) = 0$ (complete wetting, no barrier), $f(\pi/2) = \frac{1}{2}$ (hemisphere), $f(\pi) = 1$ (no wetting, the homogeneous barrier). Its derivative is $f'(\theta) = \frac{3}{4}\sin^3\theta \geq 0$ on $[0, \pi]$, so f is monotone and $0 \leq \Delta G^*_{\mathrm{het}} \leq \Delta G^*_{\mathrm{hom}}$ for every admissible θ. f is also the volume fraction of the cap relative to the full sphere of the same radius.

Assumptions. Capillarity (macroscopic interfacial energies at nucleus size), an isotropic γ_{nl} (closer to the amorphous framing than to faceted calcite), a flat, homogeneous substrate, no line tension, and a cluster of the bulk phase's composition. A non-classical pathway (prenucleation clusters, liquid precursors) is outside this framework and is a limitation, not a parameter.

Accessibility of the barrier. A barrier is low enough only relative to the number of attempts the system makes in the time available. Nuclei form on the template at the rate per unit area

$$J = J_0\,\mathrm{e}^{-\Delta G^*_{\mathrm{het}}/k_{\mathrm{B}}T} \quad [\mathrm{m}^{-2}\,\mathrm{s}^{-1}], \tag{23}$$

with J_0 the kinetic prefactor of heterogeneous nucleation per unit template area (attachment frequency, Zeldovich factor and site density). Over the time t_{p} that the template spends at the registered R-slurry state (slurry contact and drying), the expected areal density of nuclei is $N_{\mathrm{nuc}} = J\,t_{\mathrm{p}}$; along a path of registered states k held for Δt_k it is $\sum_k J_0 \Delta t_k\,\mathrm{e}^{-\Delta G^*_{\mathrm{het},k}/k_{\mathrm{B}}T}$. Requiring at least the areal density N_{req} gives the absolute-barrier criterion

$$N_{\mathrm{nuc}} \geq N_{\mathrm{req}} \iff \Delta G^*_{\mathrm{het}} \leq B_{\mathrm{acc}} = k_{\mathrm{B}}T\ln\frac{J_0\,t_{\mathrm{p}}}{N_{\mathrm{req}}} \quad [\mathrm{J}]. \tag{24}$$

Units: $J_0 t_{\mathrm{p}}$ and N_{req} are both m^{-2}, so the logarithm is of a pure number. Equivalently, a count $\mathcal{N}$ on a collector area A is $N_{\mathrm{req}} = \mathcal{N}/A$. Four properties make the criterion usable without a precise kinetic theory: only $\ln J_0$ enters, so an order-of-magnitude prefactor moves B_{acc} by a few $k_{\mathrm{B}}T$; the barrier is evaluated at Ω_{bulk} (Section 7); the non-stationary time lag is neglected; and N_{nuc} ignores exclusion zones around growing islands, so it is an upper estimate and the criterion is necessary, not sufficient. It is a statement about nucleation in the slurry, not about corrosion, and no rate of anything else follows from it.

6 Young's relation and the fixed sign convention

Physics. At the contact line, moving the line outward by a small amount replaces substrate–liquid interface by substrate–nucleus interface and stretches the nucleus–liquid surface. At equilibrium these balance. The contact angle therefore measures how much the substrate prefers the nucleus to the liquid, in units of the nucleus–liquid energy.

Convention, in words. θ is the angle at the three-phase line measured through the nucleus. $\cos\theta$ is the free energy gained per unit contact area by replacing substrate–liquid interface with substrate–nucleus interface, divided by the nucleus–liquid interfacial free energy. $\theta = 0$ means the nucleus wets the template completely; $\theta = \pi$ means it does not wet it at all.

Convention, as an equation.

$$\cos\theta = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}}{\gamma_{\mathrm{nl}}}, \qquad \gamma_{\mathrm{sn}} = \gamma_{\mathrm{sn}}^{(0)} + \Delta\gamma_{\mathrm{w}} \quad [1;\ \mathrm{J\,m^{-2}}]. \tag{25}$$

Units: a ratio of interfacial free energies. A finite angle exists only for $|\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}| \le \gamma_{\mathrm{nl}}$. For $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} \ge \gamma_{\mathrm{nl}}$ the nucleus phase spreads ($S_{\mathrm{spread}} \ge 0$, Section 8) and $f = 0$; for $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} \le -\gamma_{\mathrm{nl}}$ it does not wet and $f = 1$. An implementation clips $\cos\theta$ to $[-1, 1]$ and reports which branch applied.

The water-displacement term. The template is hydrated. When the nucleus forms on it, the template-bound water layer under the contact disc must be displaced, and that costs free energy over the contact area only. Define

$$\Delta\gamma_{\mathrm{w}} \equiv \gamma_{\mathrm{sn}} - \gamma_{\mathrm{sn}}^{(0)} \quad [\mathrm{J\,m^{-2}}], \tag{26}$$

where $\gamma_{\mathrm{sn}}^{(0)}$ is a substrate–nucleus energy from a reference that does not include the removal of that layer (for example, a contact computed without the template's structured water). The sign convention is: $\Delta\gamma_{\mathrm{w}} > 0$ **is a cost.** From Eq. (25), $\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = -1/\gamma_{\mathrm{nl}} < 0$, and with $\partial f/\partial\cos\theta = -\frac{3}{4}\sin^2\theta \le 0$,

$$\frac{\partial f}{\partial\Delta\gamma_{\mathrm{w}}} = \frac{3\sin^2\theta}{4\gamma_{\mathrm{nl}}} \ge 0 \quad [\mathrm{m^2\,J^{-1}}]: \tag{27}$$

a positive displacement cost raises θ, f and $\Delta G^*_{\mathrm{het}}$, and $\Delta\gamma_{\mathrm{w}} < 0$ lowers them. If γ_{sn} is supplied directly as a full in-water interfacial free energy, the displacement is already inside it, $\Delta\gamma_{\mathrm{w}} \equiv 0$, and the term must not be added again. Applying it twice is the double-counting error that C3 item 13 demonstrates.

Why Young uses the bare γ_{nl} and not γ_{eff}. The deck writes $\gamma_{\mathrm{eff}} = \gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}}$ and places it in the denominator of Young. That is rejected, for three reasons. (1) Geometry: the displaced water sits on the contact disc. The free surface of the cap, whose energy is $\gamma_{\mathrm{nl}}A_{\mathrm{cap}}$, never touches the template, so nothing about the template's hydration changes γ_{nl}. (2) Consistency: the factorisation of Eq. (22) requires the γ_{nl} in Young to equal the γ_{nl} that multiplies A_{cap} and sets $\Delta G^*_{\mathrm{hom}}$. A modified denominator would leave a residual surface term and break $\Delta G^*_{\mathrm{het}} = f\Delta G^*_{\mathrm{hom}}$. (3) Sign: with γ_{eff} in the denominator,

$$\left.\frac{\partial\cos\theta}{\partial\Delta\gamma_{\mathrm{w}}}\right|_{\mathrm{deck}} = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}^{(0)}}{(\gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}})^2} \quad [\mathrm{m^2\,J^{-1}}], \tag{28}$$

which is positive on the wetting side ($\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}^{(0)}$): a displacement *cost* would lower the barrier. This is design flag 3. γ_{eff} is kept only as a named diagnostic, so that W4 can reproduce the deck's form for the double-counting demonstration. The contract encodes the reversal as sign-flip test SF1 (Table 2).

Test, change	Registered convention (contract v2)	Rejected form
SF1, $\Delta\gamma_{\mathrm{w}} > 0$	$\cos\theta \downarrow$, $\Delta G^{*}_{\mathrm{het}} \geq \Delta G^{*}_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$	deck (γ_{eff} in the denominator), $\gamma_{\mathrm{sl}} > \gamma^{(0)}_{\mathrm{sn}}$: $\cos\theta \uparrow$, $\Delta G^{*}_{\mathrm{het}} \leq \Delta G^{*}_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$
SF1, $\Delta\gamma_{\mathrm{w}} < 0$	$\Delta G^{*}_{\mathrm{het}} \leq \Delta G^{*}_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$	reversed
SF2, $\Delta G_{\mathrm{seq}} \downarrow$	$\Omega = \Omega_{\mathrm{bulk}}$; $\cos\theta \downarrow$ through $\Delta\gamma_{\mathrm{ads}}$; barrier raised at fixed $\gamma^{(0)}_{\mathrm{sl}}, \gamma_{\mathrm{sn}}$	local-ideal (v1): $\Omega_{\mathrm{local}} > \Omega_{\mathrm{bulk}}$, barrier lowered
SF3, $\gamma_{\mathrm{nl}} \downarrow$	$f \downarrow$ if $\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}$; $f \uparrow$ if $\gamma_{\mathrm{sl}} < \gamma_{\mathrm{sn}}$; $f = \frac{1}{2}$ fixed if equal	"lowering γ_{nl} always lowers f"

Table 2: Sign-flip tests SF1–SF3 of `theory/contract.yaml`.

7 Where the template acts: bookkeeping and the no-double-counting proof

Physics. The template is both an interface and a ligand field. In classical nucleation the barrier depends on four quantities, $(\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}, \gamma_{\mathrm{nl}}, \Delta\mu)$; γ_{nl} belongs to the nucleus and the solution, so the template can act only through γ_{sl}, γ_{sn} and $\Delta\mu$. Section 4 showed that at equilibrium it cannot act through $\Delta\mu$. What remains is to find where its Ca^{2+} binding goes, to show that it is counted exactly once, and to state the rule the screen follows.

7.1 Two bookkeepings

Uniform chemical potential (U). Every ion carries its bulk electrochemical potential, Eq. (12); the driving force is $k_{\mathrm{B}}T \ln \Omega_{\mathrm{bulk}}$, Eq. (13); and every interaction between the template and the ions near it is an excess quantity per area inside the grand-canonical interfacial energies γ_{sl} and γ_{sn}, evaluated at the bulk chemical potentials.

Local-ideal (L). The enriched Ca^{2+} is given the bulk standard state at the local concentration, so the enrichment raises the supersaturation, $\Omega_{\mathrm{li}} = (c_{\mathrm{local}}/c_{\mathrm{bulk}})\Omega_{\mathrm{bulk}}$, and the interfacial energies are taken for the template without its Ca^{2+} excess. This was contract v1.

Which is consistent. (U) is the equilibrium statement. (L) makes a hidden assumption: an ion in the layer carries $\tilde{\mu}_{\mathrm{Ca,bulk}} - \Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$, more than the bulk ion when $\Delta G_{\mathrm{seq}} < 0$. That is the electrochemical potential of a layer that is *not* in equilibrium with the slurry, a separate reservoir held above the bulk. (L) is therefore self-contradictory: $\mathcal{E}$ is computed from the equilibrium condition (12) and then used as if that condition did not hold. The binding free energy is spent once to hold the ion at the site and credited a second time as driving force.

When a local excess of driving force is real. A genuine $\Delta\mu_{\mathrm{local}} > \Delta\mu_{\mathrm{bulk}}$ needs a non-equilibrium state with a time scale: (a) concentration during drying, when evaporation or a drying front outruns diffusive exchange with the remaining liquid; (b) slow exchange of the layer with the reservoir, for example a layer loaded with Ca^{2+} at one activity and then exposed to another, or Ca^{2+} released by protonation of the sites when the pH changes, which leaves the layer above the bulk chemical potential until diffusion relaxes it; (c) local pH excursions that shift the carbonate speciation at the interface. Each needs transport and kinetics with a stated time scale. They lie outside a thermodynamic screen, which evaluates equilibrium states at registered inputs, and P0 makes no statement about them. What a screen can do is evaluate Ω_{bulk} at each registered state along the drying path, because evaporation does change the bulk activities; that is the bulk driving force at a later state, not a local excess.

7.2 Exact reference: the grand potential of a cap

At fixed T and fixed bulk chemical potentials the open system minimises its grand potential. Forming a cap of radius r on the template transfers $n = V_{\mathrm{cap}}/v_{\mathrm{m}}$ formula units from the reservoir, each at the bulk chemical potentials, creates the cap surface and replaces template–liquid interface by template–nucleus interface over the base:

$$\Delta\Omega_{\mathrm{G}}(r) = -n\,k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}} + \gamma_{\mathrm{nl}}A_{\mathrm{cap}} + (\gamma_{\mathrm{sn}} - \gamma_{\mathrm{sl}})A_{\mathrm{base}} \quad [\mathrm{J}], \tag{29}$$

with $\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}, \gamma_{\mathrm{nl}}$ the grand-canonical (excess grand potential per area) energies at the bulk chemical potentials. Within capillarity this is exact. The reservoir term carries no template dependence, and by Gibbs' construction every effect of the template on the ions around it is an excess per area: there is no other place for it to sit. Equation (29) has the form of Eq. (19), so the factorisation of Section 5 applies with $\Delta\mu = k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}}$ and the in-situ γ_{sl}.

7.3 Gibbs adsorption: where the bound Ca^{2+} goes

Template–liquid interface. At fixed T the Gibbs adsorption equation is $\mathrm{d}\gamma_{\mathrm{sl}} = -\sum_i \Gamma_i\,\mathrm{d}\mu_i$. The Ca^{2+} binding computations use charge-neutral cells, so the adsorbing component is Ca^{2+} with its compensating charge, and the electrostatic terms cancel as in Section 4(ii). Raising the Ca^{2+} activity from zero (the Ca-free template, $\gamma_{\mathrm{sl}}^{(0)}$) to the regime value at fixed pH and ionic strength, with the Langmuir excess of Eq. (9) and $\mathrm{d}\mu_{\mathrm{Ca}} = RT\,\mathrm{d}\ln c$ (dilute bulk, fixed activity coefficient):

$$\Delta\gamma_{\mathrm{ads}} \equiv \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sl}}^{(0)} = -\int_0^{c_{\mathrm{bulk}}} \Gamma_{\mathrm{Ca}}(c)\,RT\frac{\mathrm{d}c}{c} = -RT\Big[\Gamma_{\mathrm{s}}\ln(1+x) - \delta_{\mathrm{s}}c_{\mathrm{bulk}}\Big] \quad [\mathrm{J\,m^{-2}}], \tag{30}$$

because $\int_0^{c_{\mathrm{bulk}}} \Gamma_{\mathrm{s}}\frac{x(c)}{1+x(c)}\frac{\mathrm{d}c}{c} = \Gamma_{\mathrm{s}}\ln(1+x)$ with $x(c) = \mathcal{E}c/c_{\max}$ linear in c, and $\int_0^{c_{\mathrm{bulk}}} \delta_{\mathrm{s}}c\,\frac{\mathrm{d}c}{c} = \delta_{\mathrm{s}}c_{\mathrm{bulk}}$. Units: $RT\Gamma_{\mathrm{s}}$ is $\mathrm{J\,mol^{-1}\cdot mol\,m^{-2}} = \mathrm{J\,m^{-2}}$, and $\delta_{\mathrm{s}}c_{\mathrm{bulk}}$ is $\mathrm{mol\,m^{-2}}$. In terms of the C2 inputs, $x = \exp(-\Delta G_{\mathrm{seq}}/RT)\,c_{\mathrm{bulk}}\delta_{\mathrm{s}}/\Gamma_{\mathrm{s}}$, so $\Delta\gamma_{\mathrm{ads}}$ is a function of ΔG_{seq}, $a_{Ca^{2+}}$ (through c_{bulk}), Γ_{s} and δ_{s} only. Its properties:

- $\partial\Delta\gamma_{\mathrm{ads}}/\partial\Delta G_{\mathrm{seq}} = \Gamma_{\mathrm{s}}\Theta > 0$: each site contributes its occupancy, so deeper binding lowers γ_{sl};
- $\partial\Delta\gamma_{\mathrm{ads}}/\partial\mu_{\mathrm{Ca}} = -\Gamma_{\mathrm{Ca}}$: the Gibbs equation itself;
- at strong binding ($x \gg 1$), $\Delta\gamma_{\mathrm{ads}} \to \Gamma_{\mathrm{s}}\Delta G_{\mathrm{seq}} + RT\Gamma_{\mathrm{s}}\ln(c_{\max}/c_{\mathrm{bulk}})$ plus terms that vanish: each site gives its binding free energy, less the free energy of concentrating the ion from the bulk concentration to that of a filled site;
- a template that repels Ca^{2+} ($\Delta G_{\mathrm{seq}} \to +\infty$) has $\Delta\gamma_{\mathrm{ads}} \to +RT\delta_{\mathrm{s}}c_{\mathrm{bulk}} > 0$: negative adsorption raises γ_{sl};
- with no specific interaction ($\Delta G_{\mathrm{seq}} = 0$) $\Delta\gamma_{\mathrm{ads}} = RT\Gamma_{\mathrm{s}}[y - \ln(1+y)]$ with $y = c_{\mathrm{bulk}}\delta_{\mathrm{s}}/\Gamma_{\mathrm{s}}$, a pure site-exclusion correction that vanishes as $y^2/2$ in the dilute limit.

If γ_{sl} is computed directly at the regime activities, the adsorption is already inside it and $\Delta\gamma_{\mathrm{ads}} \equiv 0$, the same rule as for $\Delta\gamma_{\mathrm{w}}$ (Section 6).

Template–nucleus interface. Under the contact disc the solution is gone. A site there is coordinated by an ion of the nucleus lattice, which came from the reservoir at the bulk chemical potential and is counted in n like every other formula unit, or it is left to carbonate or to the residual hydration layer. In either case the template–nucleus interface holds no solution-side adsorption excess: with the dividing surface at the template and a stoichiometric nucleus, γ_{sn} is the excess free energy of the template–crystal contact (with $\Delta\gamma_{\mathrm{w}}$ for the displaced water) and does not depend on μ_{Ca}. **The adsorbed Ca^{2+} belongs to the template–liquid interface only.** The template's affinity for Ca^{2+} can favour nucleation only through γ_{sn}, if its ligands coordinate lattice Ca^{2+} well; that is a property of the template–crystal contact, and it is computed or fetched as such (the C3 inputs), never inferred from ΔG_{seq}.

Consequence for the barrier (Q1). Young's relation becomes

$$\cos\theta = \frac{\gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}} - \gamma_{\mathrm{sn}}^{(0)} - \Delta\gamma_{\mathrm{w}}}{\gamma_{\mathrm{nl}}}, \qquad \frac{\partial\cos\theta}{\partial\Delta G_{\mathrm{seq}}} = \frac{\Gamma_{\mathrm{s}}\Theta}{\gamma_{\mathrm{nl}}} > 0 \quad [1;\ \mathrm{mol\,J^{-1}}]. \tag{31}$$

At fixed $\gamma_{\mathrm{sl}}^{(0)}$ and γ_{sn}, stronger binding lowers $\cos\theta$ and raises θ, f and $\Delta G^{*}_{\mathrm{het}}$. Ions held at the template–liquid interface must be stripped from the area the nucleus covers, which is a cost; the physics is that of an adsorbed inhibitor. A Ca-binding template promotes nucleation only if $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}$ stays positive after the adsorption term, that is, if γ_{sn} falls by more than γ_{sl} does. The sign of the template's effect is the sign of the difference between the Ca^{2+} excess at the template–liquid interface (here Γ_{Ca}) and at the template–nucleus interface (here zero). This is the possibility v1 flagged in Q1, now settled: bookkeeping (L) gave $\Delta G_{\mathrm{seq}} < 0$ a barrier-lowering effect it does not have (sign-flip test SF2).

7.4 The no-double-counting proof

Statement. In the screen composite

$$\Delta G^{*}_{\mathrm{het}} = f\Big(\theta\big[\gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}}(\Delta G_{\mathrm{seq}}, c_{\mathrm{bulk}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}}),\ \gamma_{\mathrm{sn}}^{(0)},\ \Delta\gamma_{\mathrm{w}}\big]\Big)\,\frac{16\pi\gamma_{\mathrm{nl}}^{3}v_{\mathrm{m}}^{2}}{3\big(k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}}\big)^{2}} \quad [\mathrm{J}] \tag{32}$$

every effect of the template on the barrier is counted exactly once.

Proof. (1) *Exactness.* Equation (29) is the free energy of cap formation in the open system within capillarity; Eq. (32) is its stationary value after the factorisation of Section 5, with $\gamma_{\mathrm{sl}} = \gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}}$. (2) *Channel separation.* Ω_{bulk} depends on bulk activities only, so $\partial\Omega/\partial(\text{any template input}) = 0$. All template dependence is in γ_{sl} and γ_{sn}, hence in θ; γ_{nl}, V_{m} and T are template-independent and may appear in both factors. (3) *Single entry of the binding.* ΔG_{seq}, Γ_{s} and δ_{s} enter γ_{sl} through Eq. (30) only, and only when γ_{sl} is built from $\gamma_{\mathrm{sl}}^{(0)}$. They do not enter γ_{sn} (template–nucleus interface, above) and do not enter Ω (Eq. (13)). The water-displacement correction enters γ_{sn} once, by the convention of Section 6. (4) Hence each template-dependent input appears once, in one interfacial energy, and only through θ. This completes the proof.

The checker verifies the syntactic form of (2) and (3) on the contract: the template inputs are partitioned into the θ channel, the Ω channel holds bulk inputs only, and ΔG_{seq}, Γ_{s}, δ_{s} reach the composite only through $\Delta\gamma_{\mathrm{ads}}$.

The two errors, demonstrated. The contract keeps two wrong composites, named and never used for a verdict (C3 item 13):

- `DG_het_local_ideal` (contract v1): Ca-free $\gamma_{\mathrm{sl}}^{(0)}$ with Ω_{li}. The binding is counted once, but in a channel that does not exist at equilibrium, and it lowers the barrier where the exact result raises it.
- `DG_het_double`: the in-situ γ_{sl} (Gibbs term inside) with Ω_{li}. The same binding free energy is charged to γ_{sl} and credited to $\Delta\mu$.

Both exceed the exact driving force by $k_{\mathrm{B}}T\ln(c_{\mathrm{local}}/c_{\mathrm{bulk}})$ per formula unit, which is $-\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$ in the dilute limit. The barrier of either is smaller than that of the exact Ω-factor by $\big[\ln\Omega_{\mathrm{bulk}}/\ln\Omega_{\mathrm{li}}\big]^{2}$, and `DG_het_double` reduces to the v1 composite without the Gibbs term and to the screen composite without the local credit (both checked). Together with the double application of $\Delta\gamma_{\mathrm{w}}$ of Section 6, these are the double-counting demonstrations.

7.5 Rule for the screen

1. The driving force is $\Delta\mu = k_{\mathrm{B}}T \ln \Omega_{\mathrm{bulk}}$ at the regime activities of Ca^{2+} and CO_3^{2-}. Neither $\mathcal{E}$, c_{local} nor Ω_{li} enters the barrier.
2. C2's ΔG_{seq} maps into the barrier only through $\Delta\gamma_{\mathrm{ads}}(\Delta G_{\mathrm{seq}}, c_{\mathrm{bulk}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}})$ of Eq. (30), added once to the Ca-free $\gamma_{\mathrm{sl}}^{(0)}$. If γ_{sl} is supplied at the regime activities, nothing is added.
3. γ_{sn} is the energy of the template–crystal contact with lattice ions at the sites (plus $\Delta\gamma_{\mathrm{w}}$ if it comes from a dry reference); it carries no term in ΔG_{seq} or $a_{\mathrm{Ca^{2+}}}$.
4. Carbonate needs no correction: a mean potential cancels in Ω and in $\Delta\mu$.
5. The same rule governs R-slurry stability: dissolution of $CaCO_3$ is decided by Ω_{bulk}. An evaluation at an "enriched" state is a non-equilibrium sensitivity, never an equilibrium result.

8 Film conformality

Physics. A barrier must be continuous: one through-path lets electrolyte reach Al. A low nucleation barrier says where nuclei form easily, not whether they spread into a film. Whether the mineral spreads or beads on the template is a wetting balance, and whether beads merge before they grow thick is a competition between how many islands form and how far each spreads.

Spreading coefficient. The free-energy change per unit area when a thick layer of the nucleus phase replaces the template–liquid interface is

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} = \gamma_{\mathrm{nl}}(\cos\theta - 1) \quad [\mathrm{J\,m^{-2}}], \tag{33}$$

the second form holding on the partial-wetting branch where Young has a solution. Units: a sum of interfacial free energies. Since $\cos\theta \le 1$, a finite contact angle always has $S_{\mathrm{spread}} \le 0$. The sign condition is: $S_{\mathrm{spread}} \ge 0$ means no finite angle exists, $\theta \to 0$, $f \to 0$, and the phase grows layer by layer (Frank–van der Merwe); $S_{\mathrm{spread}} < 0$ means $\theta > 0$ and growth by islands (Volmer–Weber). Island-then-layer growth (Stranski–Krastanov) needs a strain or misfit energy that this isotropic framework does not contain. The registered $\Delta\gamma_{\mathrm{w}}$ enters S_{spread} through γ_{sn}, with the same sign as in θ: a displacement cost makes spreading harder.

Coalescence of islands. With $S_{\mathrm{spread}} < 0$, a film forms only when islands meet. Let the islands be equal caps of contact angle θ on a square lattice of density N_{s} (spacing $\ell = N_{\mathrm{s}}^{-1/2}$). A cap of base radius ρ has height $h = \rho\tan(\theta/2)$. Neighbours touch when $2\rho = \ell$, and the base discs then cover only $\pi/4$ of the area; the square cells are covered when $\rho = \ell/\sqrt{2}$. The two geometric thicknesses are

$$h_{\mathrm{c}} = \frac{\tan(\theta/2)}{2\sqrt{N_{\mathrm{s}}}}, \qquad h_{\mathrm{cl}} = \frac{\tan(\theta/2)}{\sqrt{2N_{\mathrm{s}}}} = \sqrt{2}\,h_{\mathrm{c}} \quad [\mathrm{m}]. \tag{34}$$

Units: $N_{\mathrm{s}}^{-1/2}$ is a length. Both vanish as $\theta \to 0$ and diverge as $\theta \to \pi$. They are geometric estimates that treat caps as growing self-similarly up to contact; they are arguments, not measurements, and N_{s} comes from a nucleation-and-growth model (item 17), not from thermodynamics.

What a contact angle does and does not imply about coverage. A small θ means thin islands and early closure for a given N_{s}. It does not fix N_{s}, which depends on the nucleation rate relative to lateral growth; it says nothing about places where the template is absent, where the bare-oxide angle applies; and it is silent on porosity, on dissolution and reprecipitation of the amorphous phase, and on cracking when a hydrated film shrinks on drying. A C3 band $f(\theta) \le \frac{1}{2}$ admits θ up to $\pi/2$, which is island growth; conformality is therefore a separate criterion (C4, design flag 2).

The three C4 candidates and what each contains (lane note N1). (a) A small-angle bound $\theta \le \theta_{\max}$ and (b) $S_{\text{spread}} \ge -s$ (with $s \ge 0$ a margin) carry the same information. On the partial-wetting branch Eq. (33) gives $S_{\text{spread}} = \gamma_{\text{nl}}(\cos\theta - 1)$, so

$$S_{\text{spread}} \ge -s \iff \cos\theta \ge 1 - \frac{s}{\gamma_{\text{nl}}} \iff \theta \le \theta_{\max} = \arccos\Big(1 - \frac{s}{\gamma_{\text{nl}}}\Big), \tag{35}$$

because cos is decreasing on $[0, \pi]$ and $\gamma_{\text{nl}} > 0$. Both test one quantity, the Young numerator $\gamma_{\text{sl}} - \gamma_{\text{sn}}$ relative to γ_{nl}; they differ only in strictness. $S_{\text{spread}} \ge 0$ ($s = 0$, $\theta_{\max} = 0$) is the strict end, complete wetting; it is also the better-posed form, because it is defined when Young has no solution ($\gamma_{\text{sl}} - \gamma_{\text{sn}} > \gamma_{\text{nl}}$), where θ is not. A finite $\theta_{\max}$ needs a physical scale, and the only one the theory supplies is the coalescence condition: closure below $d_{\max}$ at island density N_{s} requires $\tan(\theta/2) \le d_{\max}\sqrt{2N_{\text{s}}}$, i.e. $\theta_{\max} = 2\arctan\big(d_{\max}\sqrt{2N_{\text{s}}}\big)$. (c) The coalescence condition $h_{\text{cl}} \le d_{\max}$, equivalently

$$N_{\text{s}} \ge N_{\text{s,cl}} = \frac{\tan^2(\theta/2)}{2d_{\max}^2} \quad [\text{m}^{-2}], \tag{36}$$

brings in N_{s}, set by nucleation relative to lateral growth, and $d_{\max}$, set by electron transport (Section 10). Neither (a) nor (b) contains either quantity, so (c) is the independent condition. C4 therefore registers one criterion from the (a)/(b) family and uses (c) as the independent supporting condition (`theory/thresholds.md`).

9 Thermodynamic stability

Physics. A layer that is thermodynamically unstable in its own environment dissolves or reacts. The environment differs between regimes, so stability is two questions: in the slurry, whether $CaCO_3$ and the native oxide survive alkaline water at the regime pH and potential; in the cell, whether the phases survive the Li reservoir at the collector's potential. Both are screens: necessary, not sufficient, and blind to morphology and kinetics.

R-slurry: aqueous Pourbaix. Each species ϕ (solid or dissolved) is brought to a common reference by exchanging O and H with water, protons and electrons at the regime pH and potential. Per non-O, non-H atom,

$$\Psi_\phi = \frac{1}{n_{\text{M}}}\Big[G_\phi - n_{\text{O}}\mu_{\text{H}_2\text{O}} - RT\ln 10\,(2n_{\text{O}} - n_{\text{H}})\,\text{pH} \\ - (2n_{\text{O}} - n_{\text{H}} + Q)FE_{\text{SHE}}\Big] \quad [\text{J mol}^{-1}], \qquad \Delta G_{\text{pbx}} = \Psi_\phi - \Psi_{\text{hull}} \ge 0. \tag{37}$$

Units: G_ϕ and $\mu_{\text{H}_2\text{O}}$ are J mol^{-1}, $RT\ln 10\,\text{pH}$ is J mol^{-1}, and FE_{SHE} is $\text{C mol}^{-1}\cdot\text{V} = \text{J mol}^{-1}$. Species with $2n_{\text{O}} - n_{\text{H}} + Q > 0$ are favoured by raising E_{SHE}, and species with $2n_{\text{O}} - n_{\text{H}} > 0$ by raising pH. Ψ_{hull} is the lower envelope over all species and mixtures of the same metal composition, and $\Delta G_{\text{pbx}} = 0$ means stable. *What it can say:* whether the phase is the equilibrium state at the chosen (pH, E_{SHE}) and dissolved-ion activities. *What it cannot say:* rates, passivation, local pH at the interface, morphology. Two traps: the amorphous framing lies above the calcite hull by its offset, so ACC is never on the hull and the tolerance must be registered (C5 open item); and the dissolved-ion activities chosen for the aqueous species decide whether solids appear stable at all, so the screen can be satisfied by construction unless it is shown to fail where it must (C5 item 21, Q8).

R-cell: Li-open grand potential. In the cell the solvent is not water and the reference is Li^+/Li. Holding the collector at E_{Li} fixes the Li chemical potential,

$$\mu_{\text{Li}} = \mu_{\text{Li}}^\circ - FE_{\text{Li}} \quad [\text{J mol}^{-1}], \qquad \Phi_\phi = G_\phi - n_{\text{Li}}\mu_{\text{Li}} \quad [\text{J mol}^{-1}]. \tag{38}$$

For a reaction $\phi \to \sum_k \nu_k \psi_k + \Delta n\,\mathrm{Li}$ that releases Δn Li to the reservoir, Li conservation gives $\Delta\Phi(E_{\mathrm{Li}}) = \sum_k \nu_k \Phi_k - \Phi_\phi = \Delta G_{\mathrm{r}} - \Delta n\, F E_{\mathrm{Li}}$, with ΔG_{r} the reaction free energy against Li metal. The phase is stable against that reaction while $\Delta\Phi > 0$, so each reaction sets one limit,

$$E^* = \frac{\Delta G_{\mathrm{r}}}{\Delta n\, F} \quad [\mathrm{V}], \qquad \begin{cases} \Delta n > 0 \text{ (delithiation, oxidation)}: & E_{\mathrm{Li}} < E^* \;\to\; E_{\mathrm{ox}} = \min E^*, \\ \Delta n < 0 \text{ (lithiation, reduction)}: & E_{\mathrm{Li}} > E^* \;\to\; E_{\mathrm{red}} = \max E^*. \end{cases} \tag{39}$$

Units: $\mathrm{J\,mol^{-1}}$ over $\mathrm{C\,mol^{-1}}$ is V. *What it can say:* whether a phase reacts with Li at the collector's potential. *What it cannot say:* oxidation that exchanges no Li. A Li-free phase such as $CaCO_3$ has no reaction with $\Delta n > 0$, so its Li-open E_{ox} is unbounded: the construction declares it stable at any anodic potential by construction. Anodic stability of $CaCO_3$ and of the Al phases needs oxidation reactions that release CO_2, O_2 or dissolved Ca^{2+} and Al^{3+} into the electrolyte, which require the electrolyte species as additional open components (Q9). Kinetics and electrolyte oxidation are outside both screens.

The oxidation sink is fixed by the electrolyte (Q9). When the collector oxidises $CaCO_3$, the carbonate gives up its electrons and leaves as CO_2 and O_2. The Ca^{2+} must go somewhere, and the cell offers only what the electrolyte contains: Li^+, the salt anion A^- and the solvent. Electroneutrality pairs each released Ca^{2+} with two A^-, either as the solid salt CaA_2 or as solvated ions, and the electrons reach the Li^+/Li reference as Li plated from the salt LiA. Referenced to Li^+/Li, the reactions are

$$\mathrm{CaCO_3} + 2\,\mathrm{LiA} \to \mathrm{CaA_2} + \mathrm{CO_2} + \tfrac{1}{2}\mathrm{O_2} + 2\,\mathrm{Li}, \qquad \mathrm{Al_2O_3} + 6\,\mathrm{LiA} \to 2\,\mathrm{AlA_3} + \tfrac{3}{2}\mathrm{O_2} + 6\,\mathrm{Li}, \tag{40}$$

each with $\Delta n > 0$ and a limit $E^* = \Delta G_{\mathrm{r}}/(\Delta n\, F)$ from Eq. (39). The sink is therefore not a free choice: it is the anion of the registered electrolyte (`charter/regimes.yaml`, `electrolyte_composition`). The rule for choosing among candidate salts, once `lit/params.yaml` lists the fluorine-free salts: (1) only the anions of the registered salt or salts are admissible; (2) the oxidation limit is the minimum over the admissible sinks and over the products each can form, because a phase is unstable if any accessible reaction is downhill; (3) the true product is the lower-free-energy of the solid salt and the solvated ions, so a limit computed with the solid salt alone is an upper bound on the true onset, and it is reported as conditional on that sink; (4) a sink whose thermochemistry is missing makes the limit *undetermined*, which never counts as a pass; (5) where the CO_2 (or O_2) reference is uncertain, the limit is evaluated under each registered reference scheme and the lowest is used. The electrolyte's own anodic limit is a separate quantity: above it the electrolyte reacts first, and it is not assessed here.

10 The insulating-film limitation

Physics. $CaCO_3$ is an electronic insulator. The film lies between the cathode coating and the Al collector, and every electron of the cathode current must cross it. A film that is continuous but too thick blocks the collector, so conformality cannot be bought with thickness. Two transport limits bound the admissible thickness; the numbers come later, through macros.

Ohmic limit. If electrons cross by bulk conduction with resistivity ρ_{e}, the area-specific resistance and the bound set by an admissible $R_{A,\max}$ are

$$R_A = \rho_{\mathrm{e}} d \quad [\Omega\,\mathrm{m^2}], \qquad d \le d^{\mathrm{ohm}}_{\max} = \frac{R_{A,\max}}{\rho_{\mathrm{e}}} \quad [\mathrm{m}]. \tag{41}$$

Units: $\Omega\,\mathrm{m}$ times m is $\Omega\,\mathrm{m^2}$; $\Omega\,\mathrm{m^2}$ over $\Omega\,\mathrm{m}$ is m.

Tunnelling limit. For a thin film the current is carried by direct tunnelling through a barrier of height ϕ_{B}. In the WKB approximation for a rectangular barrier the transmission decays as $\mathrm{e}^{-2\kappa d}$, so

$$j = j_0\,\mathrm{e}^{-2\kappa d}\quad [\mathrm{A\,m^{-2}}], \qquad \kappa = \frac{\sqrt{2 m_{\mathrm{e}} \phi_{\mathrm{B}}}}{\hbar}\quad [\mathrm{m^{-1}}], \qquad d \le d^{\mathrm{tun}}_{\max} = \frac{\ln(j_0/j_{\mathrm{req}})}{2\kappa}\quad [\mathrm{m}]. \tag{42}$$

Units: $\sqrt{\mathrm{kg}\cdot\mathrm{J}}/(\mathrm{J\,s}) = \mathrm{m^{-1}}$; the logarithm is of a ratio of current densities. The bound is positive only for $j_0 > j_{\mathrm{req}}$. The prefactor j_0, the effective mass and the barrier height are inputs with large uncertainty (Q10), and defects or hopping can only raise the current, so Eq. (42) is a conservative bound for a defect-free film.

Combined with conformality. The film must close before it exceeds the admissible thickness:

$$h_{\mathrm{cl}} = \frac{\tan(\theta/2)}{\sqrt{2N_{\mathrm{s}}}} \le d_{\max} = \max\left(d^{\mathrm{ohm}}_{\max},\, d^{\mathrm{tun}}_{\max}\right)\quad [\mathrm{m}]. \tag{43}$$

This is a stated limitation (C4 item 18), not a performance figure. It contains no statement about cell resistance, rate or capacity.

11 Summary: links and the inputs they consume

Physics. Each link reduces to a small set of inputs. Listing them once is what lets the parameter table (W1) fetch every input and the implementation (W4) consume every one, with nothing typed and nothing missing. Table 3 is that list; the ids are those of `theory/symbols.yaml`, and the functions are those of `theory/contract.yaml`.

Link	Regime	Contract functions	Inputs consumed (symbol ids)	Source
C1 adhesion	R-slurry	W_adh_wet, W_adh_wet_from_pmf, W_adh_dry_fn	Delta_F_pmf, A_c; comparator gamma_o, gamma_p, gamma_op, gamma_w	computed PMF; W1 for comparators
C2 sequestration	R-slurry	Delta_G_seq_cycle, E_enr, c_max, c_local, x_bind, Theta_occ, Gamma_Ca, Delta_gamma_ads	Delta_G_site, pKa, pH, T, c_bulk, Gamma_s, delta_s, V_site	computed PMF; W1 for pKa, pH, c_bulk
Super-saturation	R-slurry	Omega_bulk, Delta_mu; demonstration: Omega_local_ideal, Delta_mu_local_ideal	a_Ca, a_CO3, K_sp, T	W1 (activities from speciation at the regime pH and ionic strength)
C3 nucleation	R-slurry	r_star, DG_hom, f_theta, DG_het, cos_theta_young, cos_theta_template, gamma_sl_insitu, theta_young, B_acc, N_nuc; audit: gamma_eff_deck, cos_theta_deck	gamma_nl, gamma_sl0 (or gamma_sl in situ), gamma_sn0, Delta_gamma_w, V_m, T, J0_het, t_proc, N_req, with Ω_{bulk} and Delta_gamma_ads	W1 priors and computed slabs; framing at G2
C4 conformality	R-slurry	S_spread, h_c, h_cl, N_s_closure	C3 energies, N_s, d_max	nucleation-and-growth model (item 17); N_nuc as an upper estimate
C4 film bound	R-cell	R_A, d_max_ohm, kappa, d_max_tun	rho_e, R_A_max, phi_B, m_e, j_0, j_req, d_film	W1; R_A_max, j_req registered at G2
C5 Pourbaix	R-slurry	Psi_pbx, Delta_G_pbx	G_phase (with the ACC offset), mu_H2O, N_O, N_H, N_M, Q_charge, pH, E_SHE, T	W1 thermochemistry; materials database
C5 window	R-cell	mu_Li, Phi_Li, E_window	G_phase, N_Li, mu_Li0, E_Li, Delta_G_r, Delta_n_Li	materials database

Table 3: Each link, the contract functions that implement it, and the inputs it consumes. Every input is fetched or computed; none is a result of the study.